

AGA KHAN UNIVERSITY EXAMINATION BOARD

HIGHER SECONDARY SCHOOL CERTIFICATE

CLASS XI EXAMINATION

APRIL/ MAY 2017

Physics Paper II

Time: 2 hours 20 minutes Marks: 55

INSTRUCTIONS

Please read the following instructions carefully.

1. Check your name and school information. Sign if it is accurate.

**I agree that this is my name and school.
Candidate's signature**

2. **RUBRIC.** There are SIXTEEN questions. Answer ALL questions. Questions 14, 15 & 16 each offer TWO choices. Attempt any ONE choice from each.
3. When answering the questions:

Read each question carefully.
Use a black pointer to write your answers. DO NOT write your answers in pencil.
Use a black pencil for diagrams. DO NOT use coloured pencils.
DO NOT use staples, paper clips, glue correcting fluid, or ink erasers.
Complete your answer in the allocated space only. DO NOT write outside the answer box.
4. The marks for the questions are shown in brackets ().
5. You may use a scientific calculator if you wish.

Q.1. (Total 2 Marks)

Write any TWO differences between accuracy and precision.

S.No.	Accuracy	Precision
1		
2		

Q.2. (Total 3 Marks)

The diameter of a cylinder is 20 cm. It is measured with the help of a Vernier calliper having least count of 0.01 cm. Calculate percentage uncertainty.

Q.3. (Total 2 Marks)

Why is the magnitude of a vector always a positive value?

Q.4.

(Total 3 Marks)

Describe the 'head to tail rule' of vector addition. In this rule, can vectors be placed in any order of succession? Why?

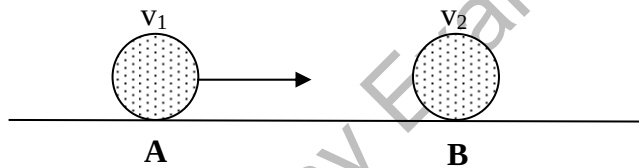
Q.5.

(Total 4 Marks)

Consider two smooth, non-rotating balls colliding elastically in an isolated system. What will be the effect on the velocity and directions of the balls if

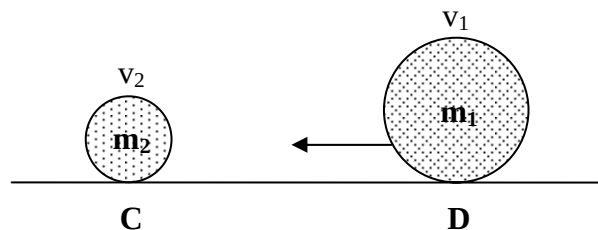
- i. ball **A**, moving in a straight line, collides with the identical, stationary ball **B** of equal mass?

(2 Marks)



- ii. heavy ball **D**, moving in a straight line, collides with a lighter ball **C** at rest?

(2 Marks)



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Q.6.

(Total 3 Marks)

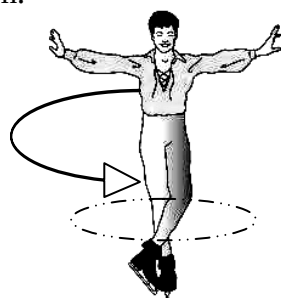
An elevator's motor lifts a mass of 85 kg to the third floor of a building at a height of 15 m with a constant velocity in 12 s. Calculate the power of the elevator's motor.

(Note: Take acceleration due to gravity ' g ' = 9.8 m/s^2 .)

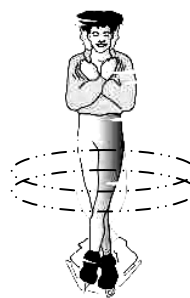
Q.7.

(Total 4 Marks)

The given figures show a skater who starts spinning and pulls his arms tightly towards his body once he gains angular momentum.



A



B

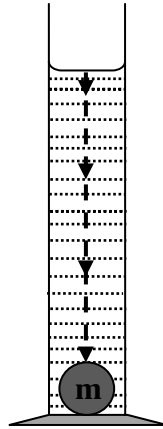
- i. State the law of conservation of angular momentum with its mathematical expression. (2 Marks)

- ii. Describe TWO changes which occur after the skater pulls his arms as stated. Give your answer with reference to the law of conservation of angular momentum. (2 Marks)

Q.8.

(Total 4 Marks)

In the given diagram, a spherical ball of mass m and radius r is falling freely in a cylinder filled with honey.



When the magnitude of the drag force F becomes zero against the weight of the spherical ball, then the ball will fall with a constant velocity v . Derive an expression for the terminal velocity of the ball using Stoke's law.

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Q.9. (Total 3 Marks)

i. Write any TWO differences between free and forced oscillations. (2 Marks)

S.No.	Free Oscillation	Forced Oscillation
1		
2		

ii. Give ONE example of forced oscillation. (1 Mark)

Q.10. (Total 2 Marks)

Define the following terms:

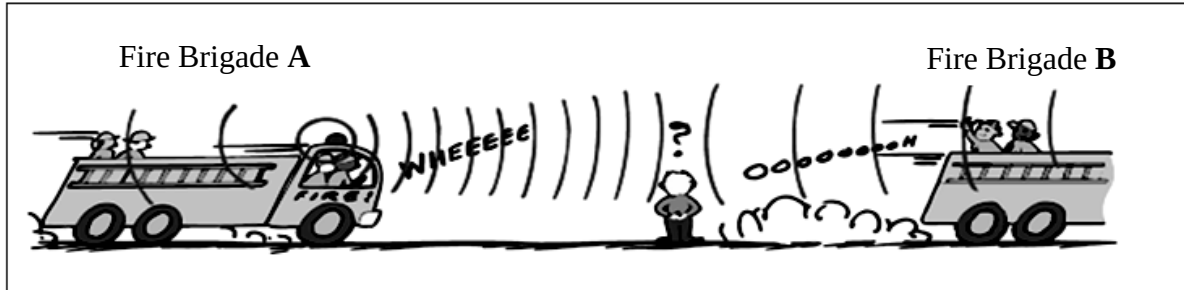
i. Oscillatory motion (1 Mark)

ii. Periodic motion (1 Mark)

Q.11.

(Total 3 Marks)

Consider two fire brigade trucks, **A** and **B**, accelerating after each other with the same speed and generating the sound of same frequency. Nearby, a listener is standing, as shown in the given diagram.



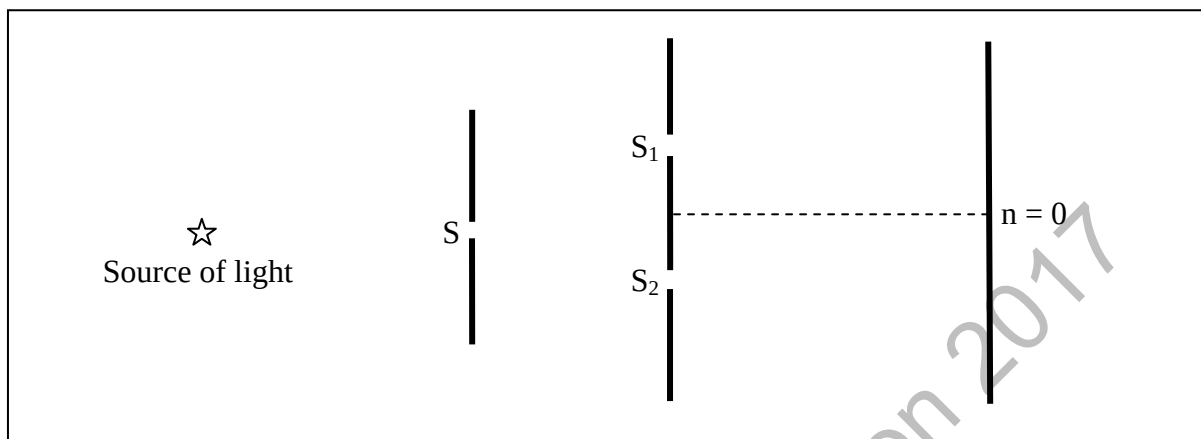
- i. To the listener, which fire brigade truck would appear to have lesser sound frequency? (1 Mark)

- ii. With reference to the stationary listener, write the mathematical expression of the apparent frequencies of sound produced by fire brigade **A** and fire brigade **B**. (2 Marks)

Q.12.

(Total 4 Marks)

- i. With the help of Hygen's principle, complete the given figure by drawing the spread of light through slits, S, S_1 and S_2 . (3 Marks)



- ii. Name the experiment used for the production and observation of the interference of light shown in the given figure. (1 Mark)

Q.13.

(Total 3 Marks)

If a Carnot engine performs 500 J of work and at the same time rejects 2000 J of heat energy to its cold reservoir, then find the percentage efficiency of the Carnot engine.

Q.14.

(Total 5 Marks)

EITHER

a.

- i. Why is it better to launch a spacecraft into its orbit from a location away from the equator? Justify your answer with the help of mathematical relation. (2 Marks)
- ii. Calculate the escape velocity of the Earth by using the given information. (3 Marks)

Mass of the Earth = 6×10^{24} kg

Radius = 6.4×10^6 m

$g = 9.8 \text{ m/s}^2$

$G = 6.67 \times 10^{-11} \text{ Nm}^2/\text{kg}^2$

OR

- b. Suppose a car of mass 1200 kg is moving with velocity of 27.5 m/s. If the engine is applying a uniform force of 3500 N, then calculate power of the car at the end of 8th second. (5 Marks)

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EITHER

- (Note:** The speed of sound in air is 340 m/s.)

OR

- (**Note:** The speed of sound in air is 340 m/s.)

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EITHER

-
- Initial Volume (dm^3)
- Final Volume (dm^3)
- Heat Source

OR

- b. Outline any FIVE environmental crises as entropy crises.

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